



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
FINAL EXAMINATION

CLASS: IX
SUBJECT: ROBOTICS & AI [SET A]

FULL MARKS:100
TIME: 2 HOURS

Instructions:

*This paper consists of 4 printed pages. This paper is divided into two sections.
Attempt all questions from Section-A and any four questions from Section-B.
The intended marks for questions or parts of questions are given in brackets[.]*

Section A
(Attempt all questions.)

Question 1

[1x10=10]

- a) Information can be defined as ____
- | | |
|---------------------|------------------------|
| i) Raw facts | ii) Abrupt sentence |
| iii) Processed data | iv) Collection of data |
- b) The three laws of robotics are created by:
- | | |
|--------------------|----------------------|
| i) Stephen Hawking | ii) Victor Scheinman |
| iii) Elon Asima | iv) Isaac Asimov |
- c) What is the advantage of using a robot in hazardous environment?
- | |
|--|
| i) They are able to operate continuously without any human intervention. |
| ii) They are less expensive |
| iii) They provide greater precision and control over a task |
| iv) They minimise the risk of human exposure to hazardous material |
- d) The _____ are the tools that are attached to the end of the manipulator.
- | | |
|-------------------|--------------|
| i) End-effectors | ii) Grippers |
| iii) Manipulators | iv) Joints |
- e) An aerial robot should contain _____
- | | |
|--|-----------------------|
| i) Light weight payload | ii) Heavy Payload |
| iii) A payload that does not affect its flying | iv) No payload at all |
- f) Which type of joint permits rotation in multiple axes?
- | | |
|----------------------|---------------------|
| i) Revolute joint | ii) Spherical joint |
| iii) Prismatic joint | iv) Rotating joint |
- g) **Assertion:** Machine learning requires good quality and sufficient data to train and test the algorithm

Reason: For correct classification, good quality data that is free from noise, and sufficient data is required for training and testing of algorithm.

- i) Assertion (A) and Reason(R) both are true and (R) is the correct explanation of (A)
 - ii) Assertion (A) and Reason(R) both are true but (R) is not the correct explanation of (A)
 - iii) Assertion is true and Reason is false
 - iv) Assertion is false and Reason is true
- h) What would be the output of the following code snippet?
- ```
def fun(x, y=2):
 return x * y
result = fun(5)
print(result)
```

- i) 5
  - ii) 10
  - iii) 2
  - iv) error
- i) Consider the following statement and answer the following question:  
**Data=({"name": "Subhash", "age": 15})**

This statement will create a-

- i) Tuple
  - ii) Set
  - iii) List
  - iv) Dictionary
- j) **Assertion (A):** Python standard library consists of number of modules  
**Reasoning (R) :** A function in a module is used to simplify the code and avoid repetitions

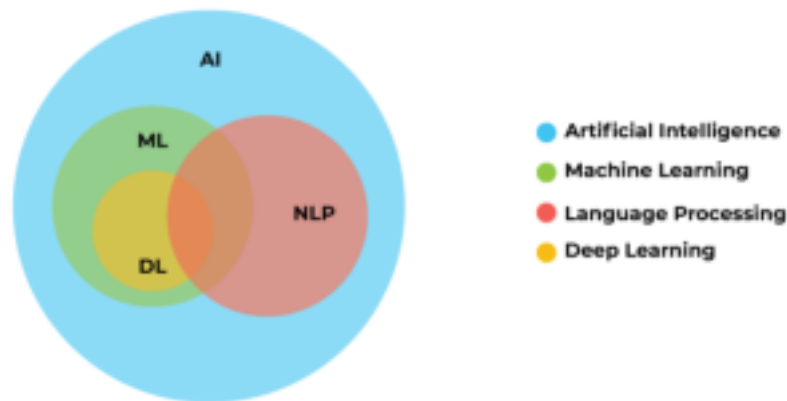
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## Question 2

- a) Correct the following python code that reverses a number. Rewrite the code and underline the corrections. [2x4=8]

```
define revNumber(num) :
 rev = 0
 rem = 0
 While num > 0:
 rem = num %10
 rev = rev*10 + rem
 num = num /10
 return rev
print(revNumber(1234))
```

b) Study the following image and answer the questions-



i) State the relationship between **AI** and **ML**. [2]

ii) What is the role of **NLP**? [2]

c) Identify the type of data in the given table [4]

| Scenario                             | Type of Data |
|--------------------------------------|--------------|
| A movie clip                         |              |
| Radio news                           |              |
| Newspaper content                    |              |
| Recording of telephonic conversation |              |

d) What is the purpose of **Turing** test with the help of an example. [2+2]

### Section – B

(Attempt any four questions from this section. Write VDT for all python programs.)

#### Question 3

a) A neon number is a **number where the sum of digits of square of the number is equal to the number**.

Example-  $9^2 = 81 = 8+1=9$

Write a program in **Python** to read a number and check if it is a **Neon** number. [14.5]

b) What are sensors? Discuss three types of sensors in brief. [3]

#### Question 4

a) A number is called an Automorphic number **if and only if its square ends in the same digits as the number itself**.

Examples-

$$5^2 = 25$$

$$6^2 = 36$$

$$76^2 = 5776$$

$$376^2 = 141376$$

Input a number from the user and write a program in **Python** to check if the number is an **Automorphic** number or not. [14.5]

b) Differentiate data with information. Give example. [3]

### Question 5

- a) Write a program in **Python** to read a number and a digit. Check if the digit **exist** in the number or not. If exists, print its **frequency**. [14.5]

Example 1. – N1 = 6426272

N2=2

Output- 2 exist in 6426272 3 times

Example 2. – N1 = 4275

N2=9

Output- 9 does not exist

- b) Discuss the three **basic** gates, draw their truth table and symbol. [3]

### Question 6

- a) ABC Diagnostic Centre announces discount on its health check-up plan as below: [14.5]

| Plan Range  | Senior Citizen (G) | General Citizen (C) |
|-------------|--------------------|---------------------|
| <1200       | 10%                | 5%                  |
| 1201 – 2000 | 20%                | 10%                 |
| >2000       | 30%                | 15%                 |

Write a program in **Python** to read the cost of the health package, name of the patient, patient type (G/C) and calculate the discount. The program should print the **patient name, patient type, package cost, discount price and amount payable**.

- b) What is **derived** gate? Give an example with its symbol and Boolean expression. [3]

### Question 7

- a) i) Write a function in **Python** that return the sum of multiples of **3** and **5** between **0** and **limit**. The limit should be read as a parameter. Example, if the limit is 20 then it should return the **sum** of 3,5,6,9,10,12,15,18,20. [8]  
ii) Print the **multiplication table** of a number entered by the user. [6.5]
- b) Name three rules of naming a variable in **Python**. [3]

### Question 8

- a) Implement the following methods in **Python**- [14.5]

a. #def prime( n) - checks if **n** is a prime number or not. **Example** if n=3 it displays “**Prime**”, if n=4 it displays “**Non Prime**”

b. #def sumOfOdds(n) - returns the sum of odd digits present in **n**. Example if **n=3281** the it will return **3+1=4**

- b) Briefly discuss three applications of **AI** in the field of **medicine/ healthcare**. [3]